

JavaScript DOM (Document Object Model) - Notes

1. What is DOM?

DOM (Document Object Model) is a programming interface for HTML and XML documents.

It represents the web page as a tree structure and allows JavaScript to manipulate content, structure, and styles dynamically.

2. DOM Tree Structure

HTML is represented as nodes in a tree:

document -> html -> body -> elements (h1, p, div, etc.)

3. Selecting Elements

- `getElementById("id")`
- `getElementsByClassName("class")`
- `getElementsByTagName("tag")`
- `querySelector("css selector")`
- `querySelectorAll("css selector")`

4. Changing Content

`element.innerText = "New Text";`

`element.innerHTML = "<b>Bold Text</b>";`

5. Changing Styles

`element.style.color = "red";`

`element.style.backgroundColor = "yellow";`

6. Working with Classes

```
element.classList.add("active");  
element.classList.remove("active");  
element.classList.toggle("active");
```

7. Events

```
element.addEventListener("click", function() {  
    alert("Button clicked!");  
});
```

8. Creating Elements

```
let p = document.createElement("p");  
p.innerText = "Hello";  
document.body.appendChild(p);
```

9. Removing Elements

```
element.remove();
```

10. Input Values

```
let value = document.getElementById("input").value;
```

11. Looping Over Elements

```
document.querySelectorAll(".item").forEach(item => {  
    item.style.color = "blue";  
});
```

12. Common Mistakes

- JS runs before HTML loads
- Wrong id/class names
- Using # with getElementById